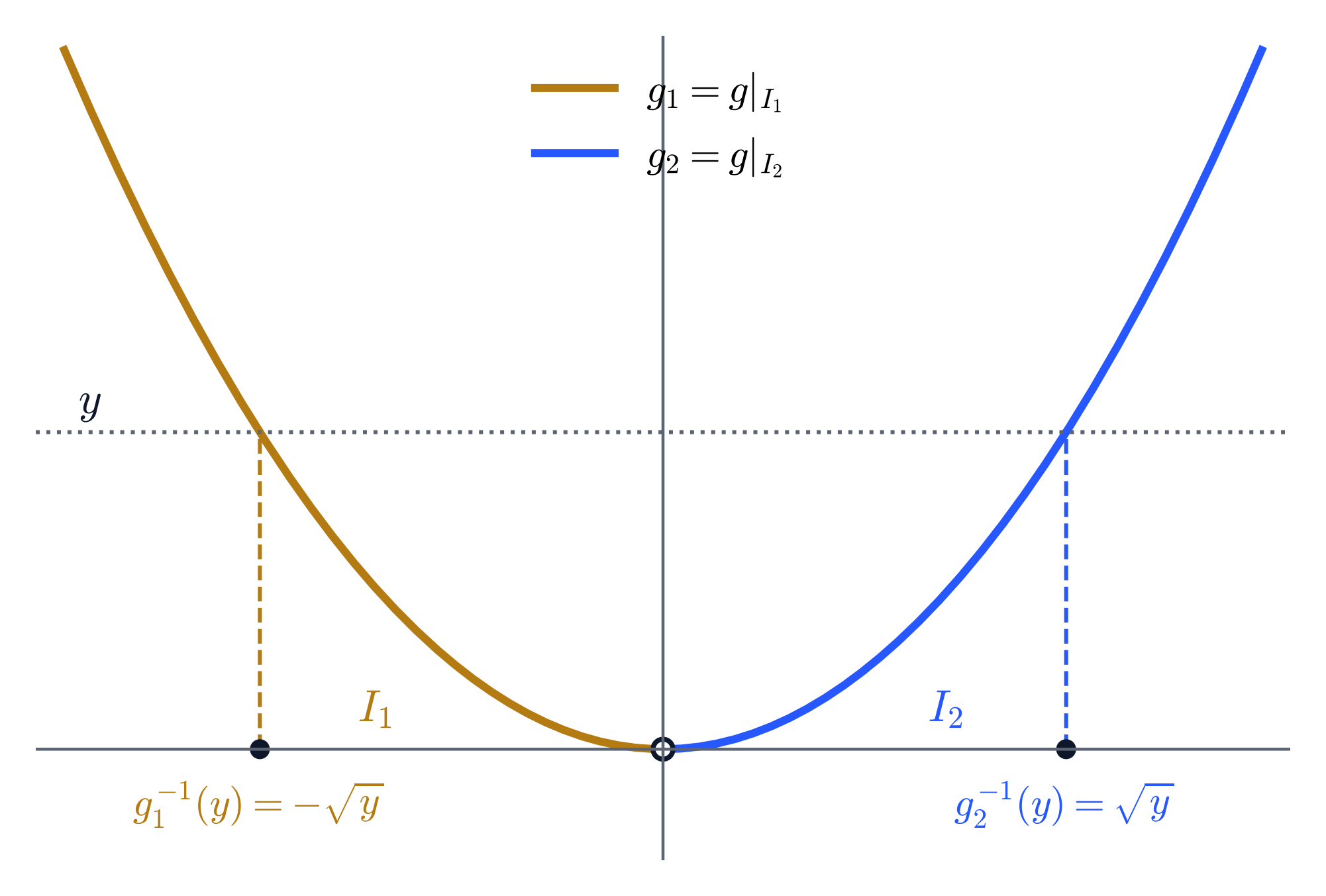
$g_1 = g|_{I_1}$
$g_2 = g|_{I_2}$
y
I_1
I_2
$g_1^{-1}(y) = -\sqrt{y}$
$g_2^{-1}(y) = \sqrt{y}$